Computer Laboratory Exercises in Optimization 2025
Lab 1: Line search and unconstrained optimization

The lab is run with the MATLAB package, version R2022a or later, which you can install freely on your computer; see `http://program.ddg.lth.se/en/`. *Make sure you choose to install the Optimization and the Symbolic Math toolboxes.* You should work in a group of 2–3 students. If you plan to do the programming project in the course with Python, you may implement the preparatory line-search algorithms in Python (but you still need to download MATLAB).

Preparations to do before the lab session in order to pass

- If you are not familiar with MATLAB you can examine this interactive program with the command `demo`. A faster way may be to type `help demos` and choose a demo from the list, e.g., type `intro`. You might find it useful to look at the MATLAB links on the course Canvas web page under Assignments.
- Read in the textbook and be sure that you understand the line-search methods and the multidimensional methods Steepest Descent, Newton and Modified Newton.
- Solve Exercises 3.6, 7 and 9 in the textbook by implementing the line-search methods Dichotomous, Golden Section, Bisection and Newton in either MATLAB or Python. Write one m-function file per method, for example, starting like (in MATLAB)

```
function [X,N]=goldensection(F,a,b,tol)
% For example F = @(x) 1-x*exp(-x)
% [a,b] interval
% tol = maximum final interval length
% X output matrix containing final a, b and b-a from every iteration
% N = number of function evaluations
```

- Download the files `Lab1a.m`, `Lab1b.m`, `Function3.m` and `Function4.m` from the course web page.
- Type `Lab1a` in the MATLAB window to start the program. A new window appears. Select 'Golden'. The function shown (`Function1`) should be minimized on the interval $-0.5 \leq x \leq 0.5$. Select an initial interval by clicking two points in the window. For each next click, anywhere in the window, a new smaller interval of uncertainty will be shown. Change the visible interval to $-10 \leq x \leq 10$. Try several starting intervals. Investigate also `Function2` on the interval $-10 \leq x \leq 10$.

During the laboratory session

1. Start MATLAB. Be ready to present your implemented line-search programs and to test them with a function given by the teacher.
2. Start `Lab1a`. Repeat the investigations (made above for Golden Section) for Newton's method. Try different initial points all over the interval $-10 \leq x \leq 10$ for both functions.
3. Questions to be answered for Golden Section and Newton's method:
   - Does the method always converge?
   - What kind of points are found? Global or local minimum, or something else?
   - Would you say that it is a reliable method?

- How far away from a minimum can you start without losing the convergence to it?

4. Investigate Armijo's method. Choose `Function1` and the interval $-0.5 \leq x \leq 0.5$. A single click in the window selects the point $\boldsymbol{x}_0$, then some possible iterations that *may occur*. What is $\boldsymbol{d}$ in the figure? Can you point out the λ-axis and its origin? Interpret and explain the red and blue asterisks. Use the sliders and explain what happens.

5. Try two MATLAB methods for minimization of `Function1` by typing

   ```
   f1=@(x) (-x^4+6*x^2)/(1+(x-0.5)^4)
   options=optimset('display','iter');
   fminbnd(f1,-1,1,options)   (the interval is here [−1, 1]; try also [−10, 10])
   [x,fval,exitflag,output]=fminunc(f1,-1)   (initial point is −1; try other)
   ```

   Do you see any similarities with Golden Section or Newton's methods? Is the MATLAB result satisfactory? Convergence? Global minimum? Do the methods use derivatives?

6. Start `Lab1b`. In a new window you can minimize functions of two variables with

   Steepest Descent with Armijo's line search

   Newton's method without line search

   Modified Newton's method without line search.

   Investigate first Steepest Descent on `Function1`. Choose an initial point with the mouse and continue clicking to see the iterations. After you have examined the behaviour for many different starting points, try Newton's method in the same way (how do you choose this method?), and finally the modified Newton for different values of ε. Investigate also `Function2` in the same way.

7. Questions to be answered for each of the three methods:

   - Does the method always converge? Is it fast or slow? For what kind of functions does it work best?
   - What kind of points are found by the method? Global or local minimizer or something else?
   - How does the size of ε in the Modified Newton method affect the performance?
   - What are the connections between the three methods?

8. For `Function2`, try to find a good strategy on how to choose a method and an ε on every iteration to find the minimizer with a few number of iterations as possible.

9. Do the same investigations for `Function3` and `Function4` first without seeing any contours (`Levels = 2`). Can you be sure that you have found the global minimum? To which points do the methods converge? Later choose more contours with the bar.

10. Download `Function3.m` and `Function4.m` from the course web page. Compare with MATLAB's `fminsearch` by typing (`X`=3 or 4)

    ```
    [point,val]=fminsearch('FunctionX',[5 5],options)
    ```

    Here [5 5] is the initial point (try several different).

    Do they find the global minimizer? Which method does MATLAB use when there is no information about the differentiability of the function? Run the appropriate downloaded demo with which you can see how the method works in two dimensions. Try different values on `simp_edge` (edit the file).